

Mapping AI Adoption in the US Economy

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| KEY QUESTION

01

How does the frequency and distribution of AI usage vary across different industries or job types?

02

How do Bright/Green jobs differ from non-Bright/Green jobs in AI usage?

03

How is AI predominantly used for automation or augmentation across different tasks and jobs?

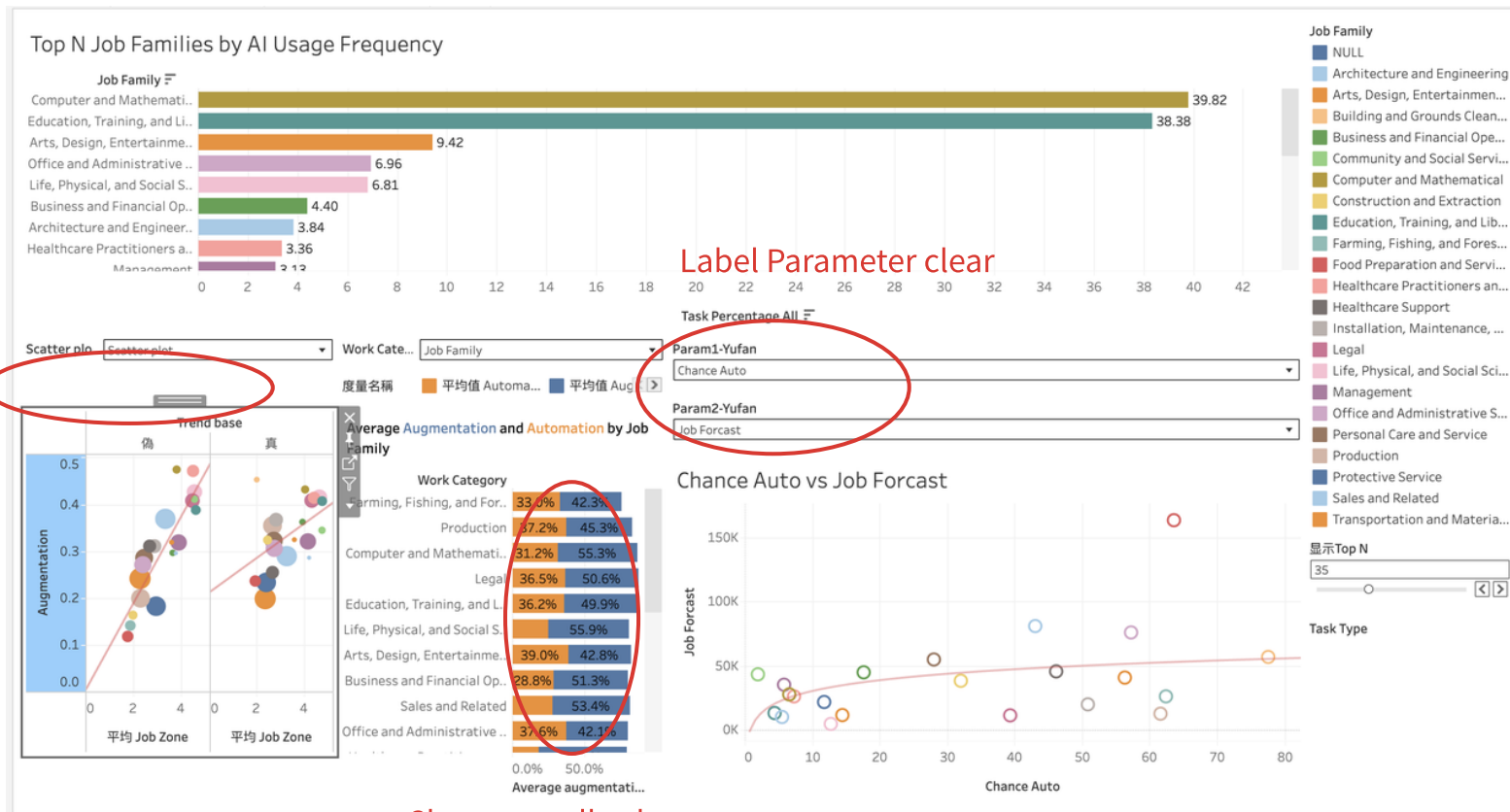
04

How does automation risk (ChanceAuto) relate to observed patterns of AI usage and augmentation/automation ratios?

IMPROVENET (Before)

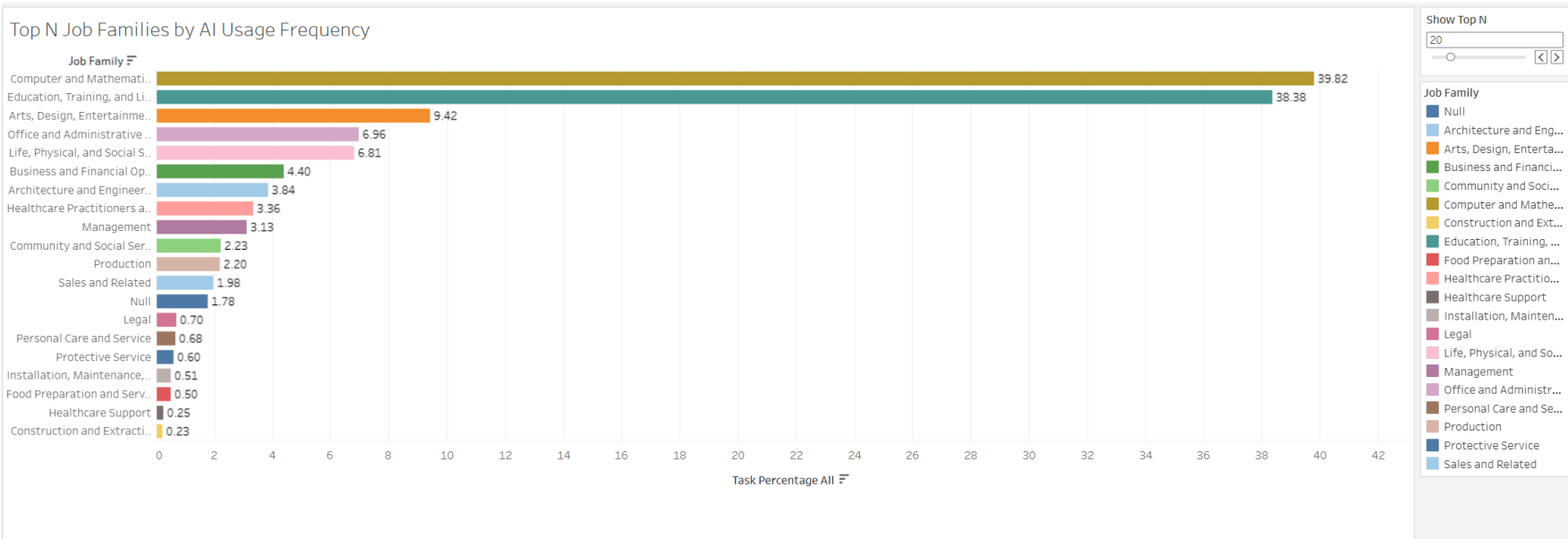
All variables are the **average** value for measurement.

Add the title of the pattern

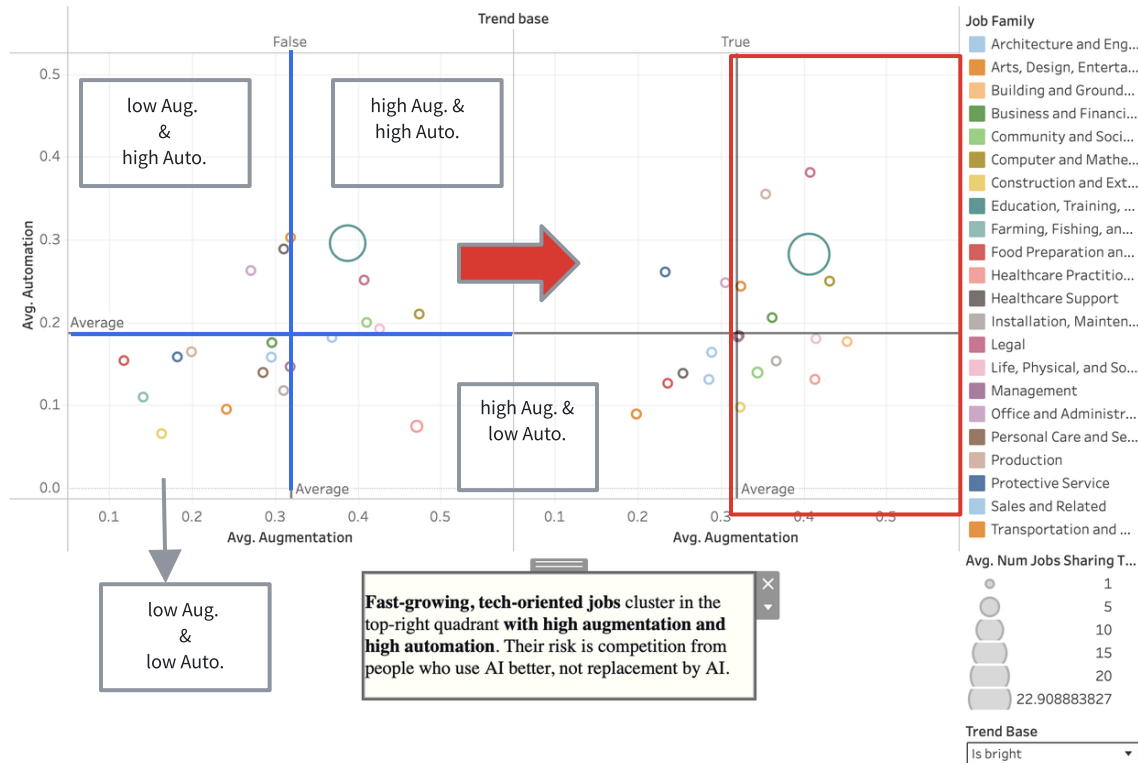


Clean up null values

DASHBOARD DESIGN-Top N Job Families by AI Usage Frequency

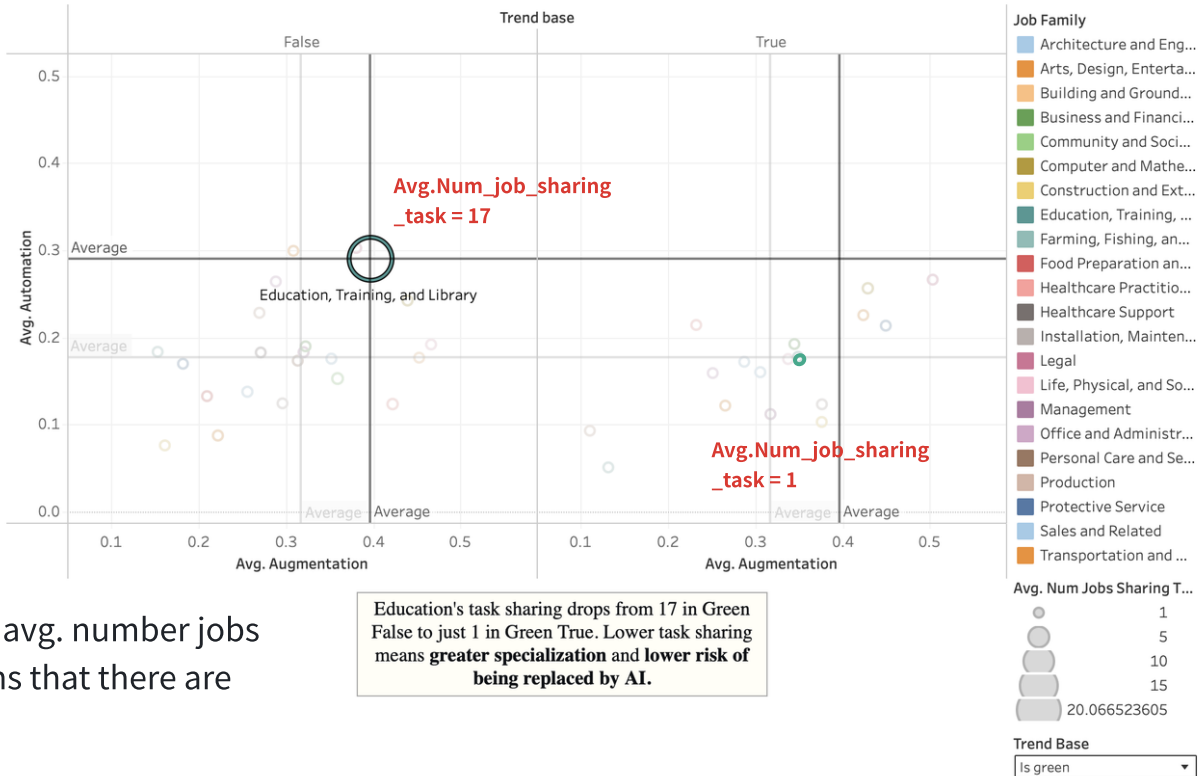


DASHBOARD DESIGN - AI usage patterns by Trend Base(Bright/ Green)



- Separated into Four quadrant
- High Aug. & high Auto.** - replaced by someone known how to use AI
- High Aug. & low Auto.** - Ideal zone (Bright outlook)
- Low Aug. & high Auto.** - easily replaced by AI
- Low Aug. & low Auto.**
- Bright Outlook jobs shift toward high augmentation.

DASHBOARD DESIGN - AI usage patterns by Trend Base(Bright/ Green)

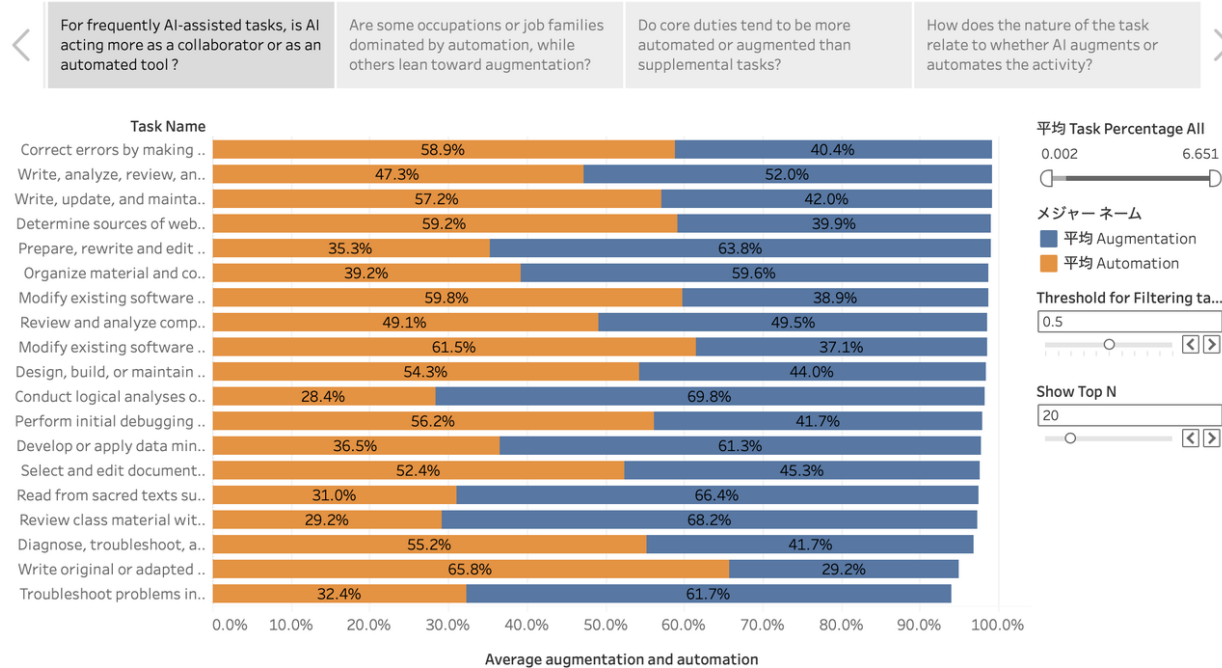


- The smaller size of avg. number jobs sharing tasks means that there are specialized tasks.

DASHBOARD DESIGN- Nature of AI Usage: Augmentation vs Automation

for frequently AI-assited tasks

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Filter tasks by removing likely **non-work activity** (Higher values mean non-work activity).

Filtered by **top-N task percentage all**.

The ratio of **augmentation** to **automation** varies depending on the **specific task**.

DASHBOARD DESIGN- Nature of AI Usage: Augmentation vs Automation

for Job Family

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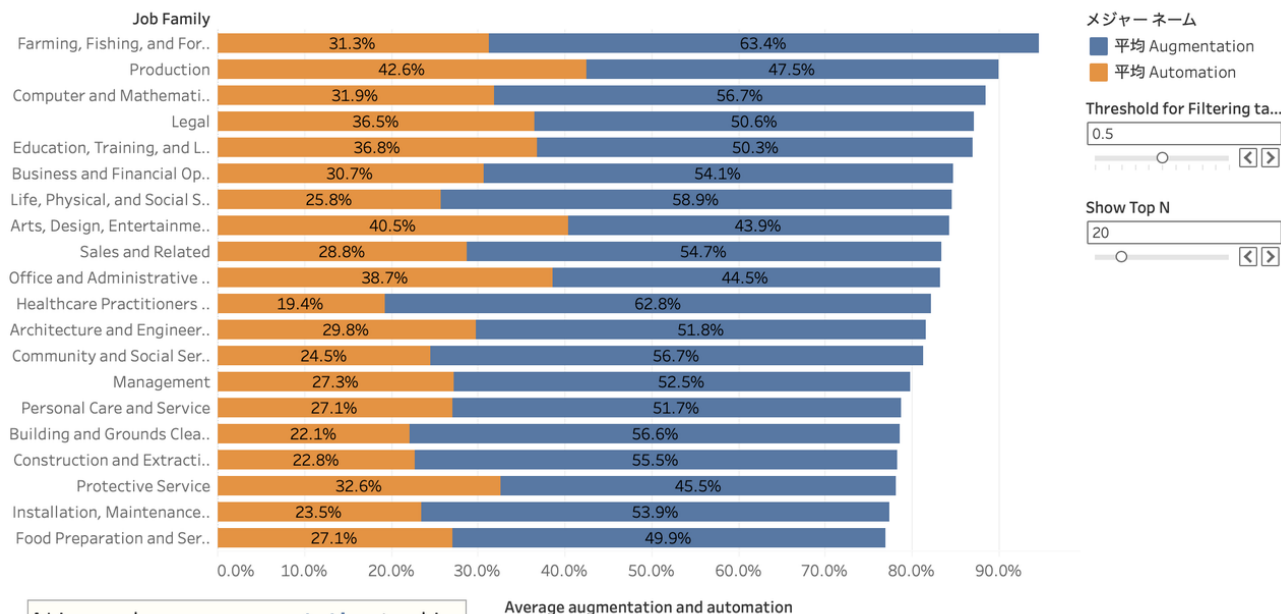
For frequently AI-assisted tasks, is AI acting more as a collaborator or as an automated tool?

Are some occupations or job families dominated by automation, while others lean toward augmentation?

Do core duties tend to be more automated or augmented than supplemental tasks?

How does the nature of the task relate to whether AI augments or automates the activity?

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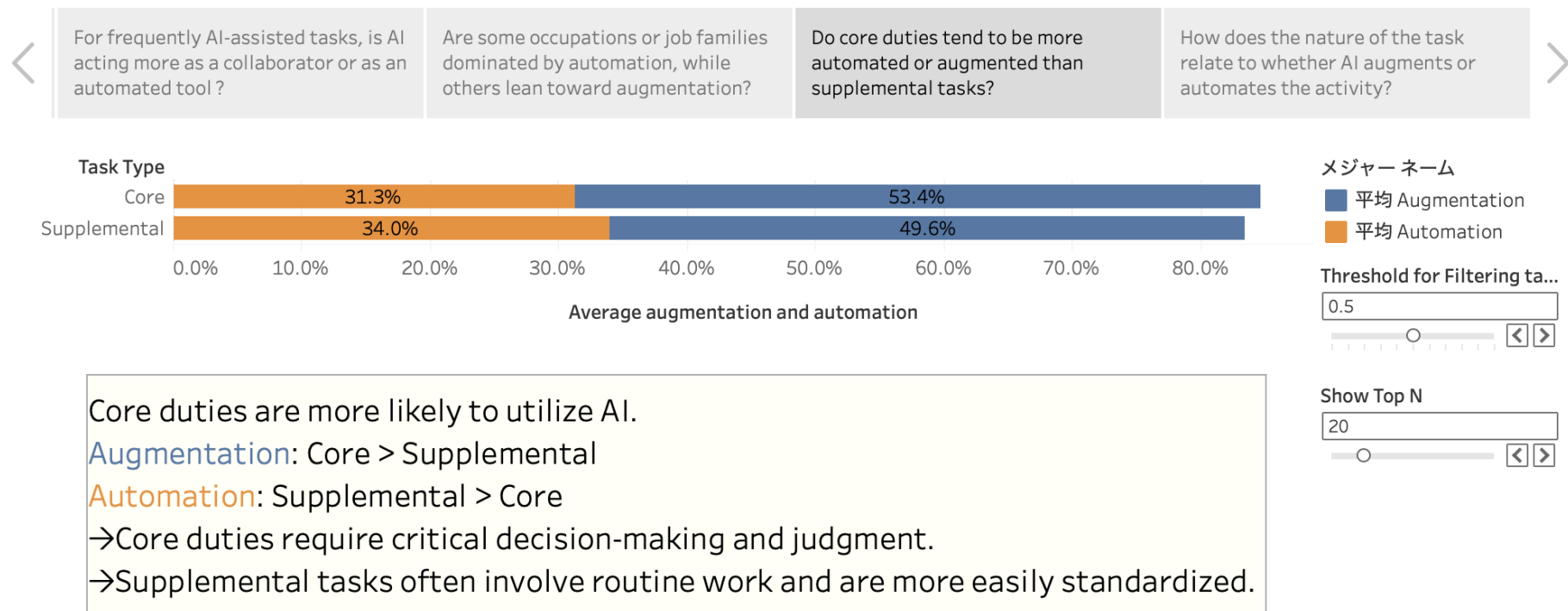


AI is used as an **augmentation** tool in all job families.

- While certain tasks can be fully outsourced to AI, entire occupations cannot yet be completely replaced.
- AI is being widely adopted across many types of work.

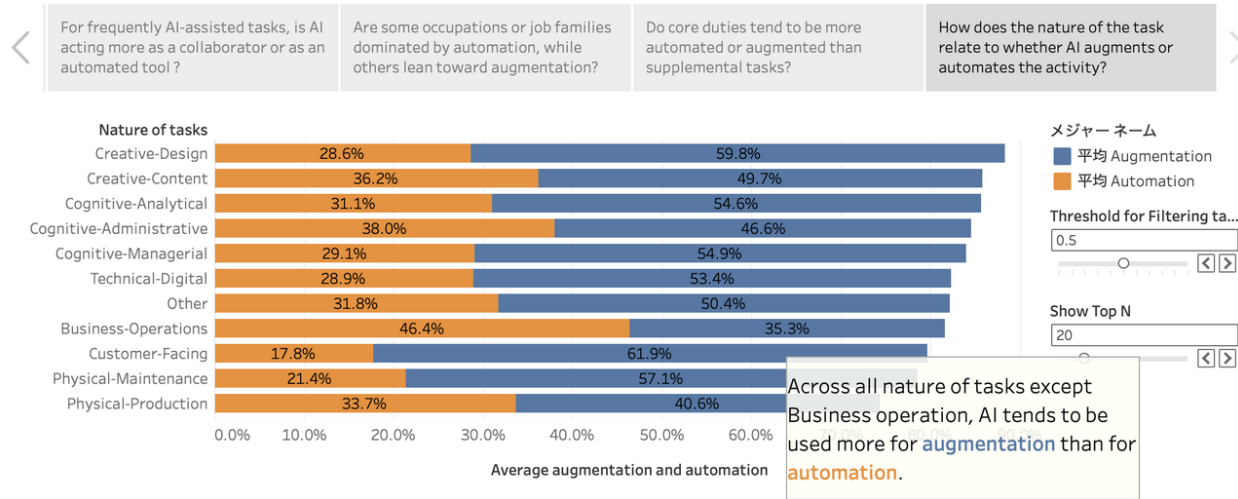
DASHBOARD DESIGN- Nature of AI Usage: Augmentation vs Automation for Task Type

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DASHBOARD DESIGN-Nature of AI Usage: Augmentation vs Automation for Nature of Tasks

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Cognitive-Analytical: A task that involves "understanding, analyzing, and evaluating" based on data and information

Cognitive-Managerial: A task that involves "planning, deciding, and controlling" organizations and operations

Cognitive-Administrative: A task that involves "processing and organizing information" according to rules

Creative-Design: A task that thinking of new structures, ideas, and designs

Creative-Content: A task that creating "expressions" such as documents and media

Technical-Digital: A task that handling systems and software

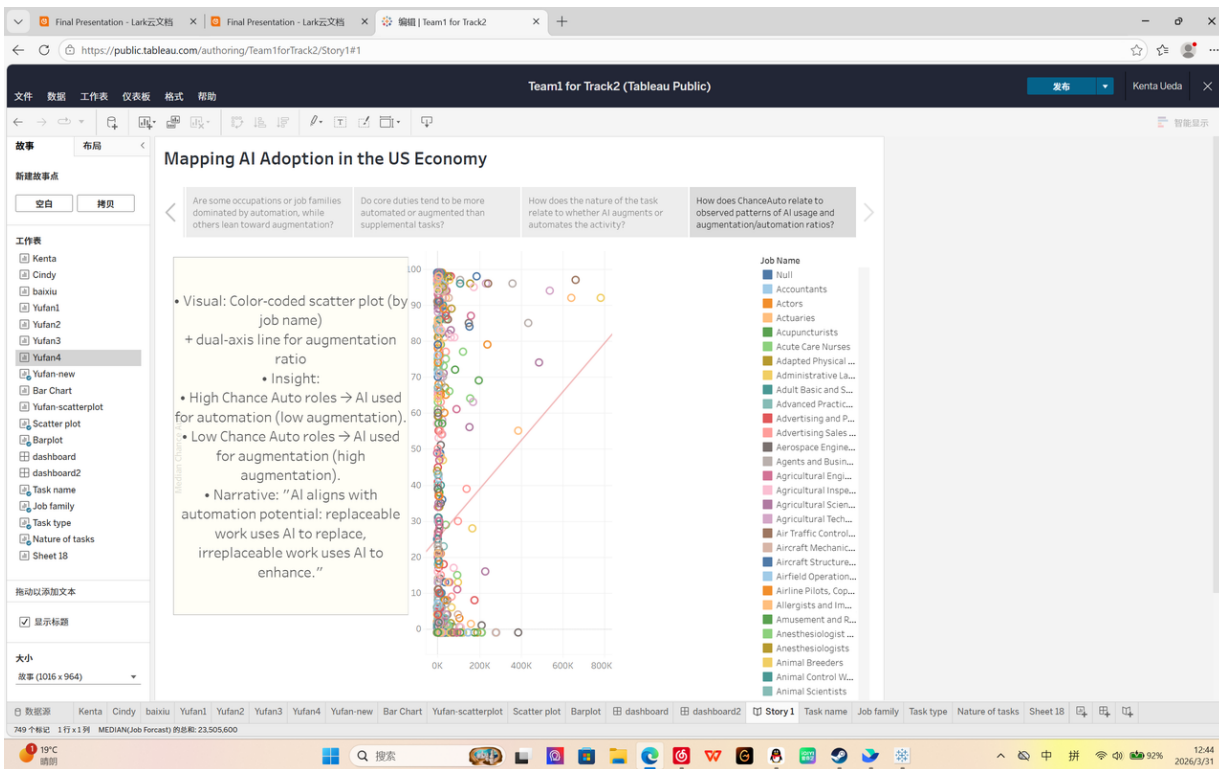
Business-Operations: A task that support the backend of business activities

Customer-Facing: Services and sales that involve direct interaction with people

Physical-Production: Work that involves making and assembling things

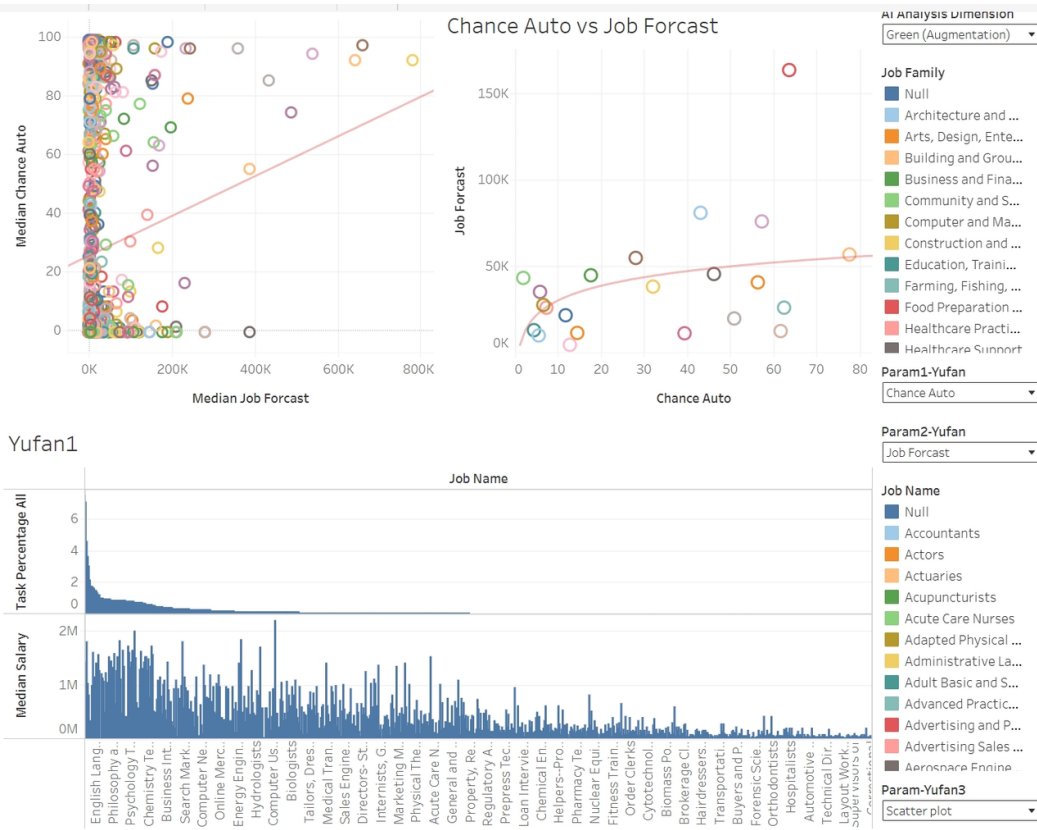
Physical-Maintenance: Work that involves maintaining and moving equipment and environments

DASHBOARD DESIGN-Nature of AI Usage:AI Use in Relation to Job Security, Wages, and Outlook



- Visual: Color-coded scatter plot (by job name)
+ dual-axis line for augmentation ratio
- Insight:
- High Chance Auto roles → AI used for automation (low augmentation).
- Low Chance Auto roles → AI used for augmentation (high augmentation).
- Narrative: "AI aligns with automation potential: replaceable work uses AI to replace, irreplaceable work uses AI to enhance."

DASHBOARD DESIGN-Nature of AI Usage:AI Use in Relation to Job Security, Wages, and Outlook



1. There is a weak positive correlation between job forecast size and automation risk (Chance Auto), but job type/industry is a much stronger factor.
2. Jobs with high automation risk use AI mainly for automation (low augmentation ratio), while jobs with low automation risk use AI to augment human work (high augmentation ratio).
3. Wage and task patterns also differ: higher-wage jobs often show lower automation risk and more AI-augmented work.

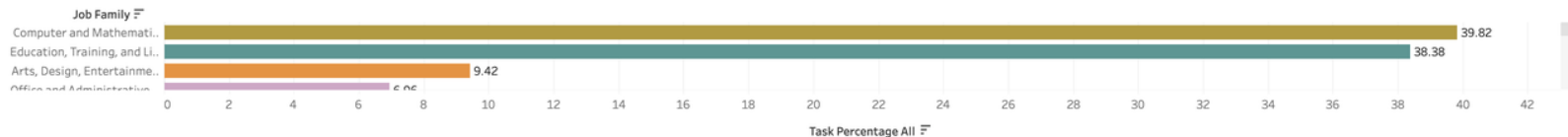
IMPROVEMENT (After)

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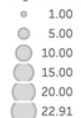
Show Top N

20

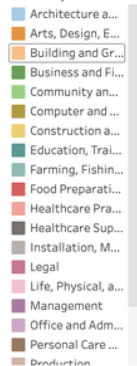
Top N Job Families by AI Usage Frequency



Avg. Num Jobs Sharin...



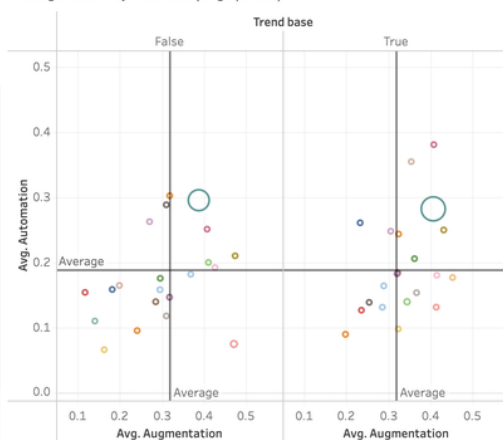
Job Family



Trend Base

Is bright

AI usage Pattern by trend base(Bright/Green)



Work Category

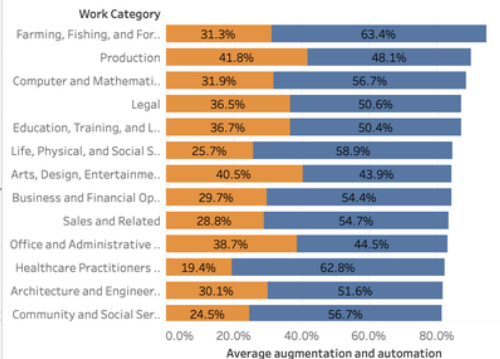
Job Family

Threshold ...

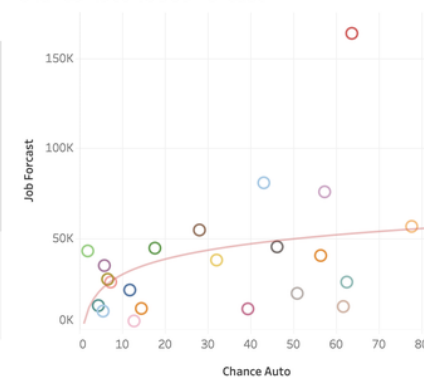
0.5

Measure N... Avg. Automation Avg. Augmenta...

How is AI predominantly used for automation or augmentation across Job Family?



Chance Auto vs Job Forecast



| CONCLUSION

AI is not simply "eliminating" jobs, but is **"reshaping" tasks**.

Currently, AI plays **an uneven but critical role** in the U.S. economy: it functions primarily as an amplifying force, especially in high-intellect, high-growth sectors. Future career resilience will no longer depend on avoiding AI, but on leveraging AI as a multiplier of professional competitiveness through specialization and by enhancing the ability to **collaborate with AI**.

| LESSONS LEARNED

- **Use consistent metrics**
 - Avoid mixing mean, median, sum, etc. to keep graphs consistent
- **Choose charts based on relationships**
 - Scatter (relationship), Bar (comparison), Line (trend over time)
- **Add clear titles to each chart**
 - Make the purpose immediately understandable
- **Label parameters clearly**
 - Use meaningful and descriptive names
- **Use the same color in the category**
 - Maintain consistency in the category colors used across different charts

Thank you

Track 2

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